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University of Asia Pacific

Department of Computer Science and Engineering

CSE 438: Robotics Lab

LAB REPORT

Experiment Number: 04

Experiment Title: Designing and Implementing a Self-Balancing Robot Using Arduino UNO

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1. Experiment Name

Designing and Implementing a Self-Balancing Robot Using Arduino UNO

2. Objective

The objective of this experiment is to **design and implement a robotic system using Arduino UNO, HC-SR04 ultrasonic sensor, L293D motor driver, and two DC motors**.
The system is designed to detect obstacles using the ultrasonic sensor and control the motors according to the detected distance.

3. Apparatus / Hardware & Software Requirements

- List all required tools and components:

Hardware Requirements:

- Arduino UNO
- L293D Motor Driver IC
- Ultrasonic Sensor
- 2 × DC Gear Motors
- Breadboard
- Jumper Wires
- 9V Battery / External Power Supply
- Robot Chassis
- Wheels

Software Requirements

- Arduino IDE
- Tinkercad Circuits (for simulation/design)
- Arduino programming language (C/C++)

4. Circuit Diagram / Schematic

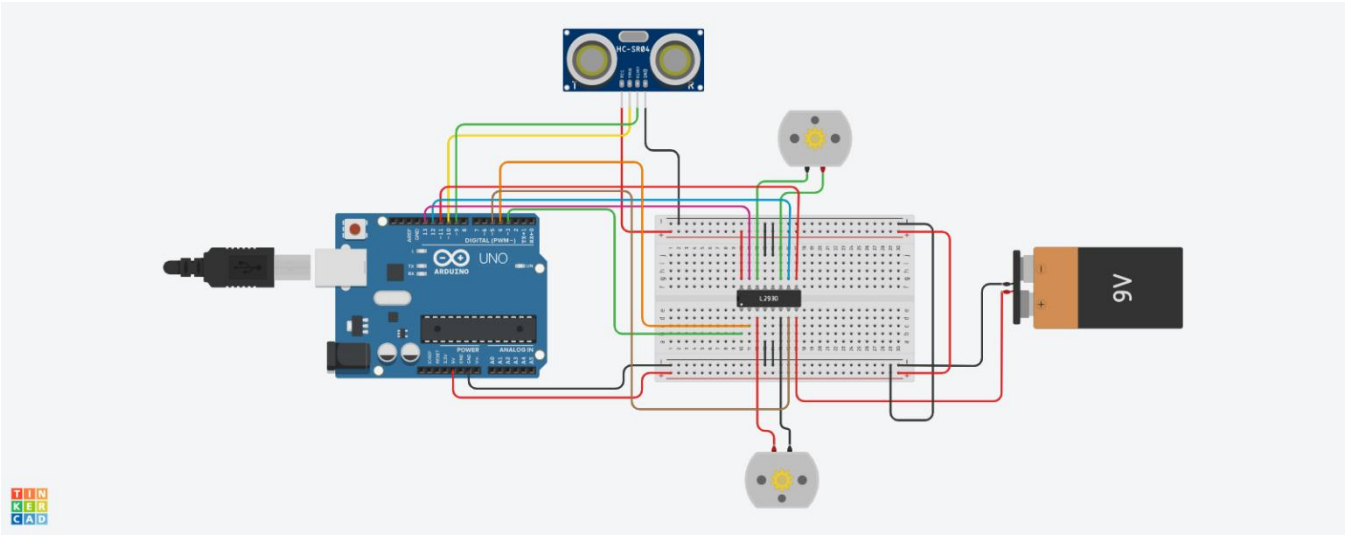


Fig: Self-Balancing Robot

5. Code / Assembly Program

```
#include<Servo.h>
int trigPin = 10;
int echoPin = 9;
long time;
float distance;
// left motor
int leftMotorSpeedPin = 3;
int leftMotorForwardPin = 4;
int leftMotorBackwardPin = 5;
// right motor
int rightMotorSpeedPin = 11;
int rightMotorForwardPin = 12;
int rightMotorBackwardPin = 13;
void setup() {
  pinMode(10, OUTPUT);
  pinMode(9, INPUT);
  Serial.begin(9600);
  pinMode(leftMotorSpeedPin, OUTPUT);
  pinMode(leftMotorForwardPin, OUTPUT);
  pinMode(leftMotorBackwardPin, OUTPUT);
  pinMode(rightMotorSpeedPin, OUTPUT);
  pinMode(rightMotorForwardPin, OUTPUT);
  pinMode(rightMotorBackwardPin, OUTPUT);
  Serial.begin(9600);
  digitalWrite(leftMotorSpeedPin, HIGH);
  digitalWrite(rightMotorSpeedPin, HIGH);
```

```
}
void loop() {
    digitalWrite(10, LOW);
    delayMicroseconds(2);
    digitalWrite(10, HIGH);
    delayMicroseconds(10);
    digitalWrite(10, LOW);
    time = pulseIn(9, HIGH);
    distance = time*0.034/2;
    Serial.print("Dsitance: ");
    Serial.println(distance);
    delay(100);
  if (distance > -1 && distance < 25) {
    stop();
    delay(100);
    goBackward();
    delay(100);
    stop();
    delay(100);
    if(random(0, 2) == 0) {
      goLeft();
    } else {
      goRight();
    }
    delay(100);
    stop();
```

```
    delay(100);
  } else {
    goForward();
  }
}
void goForward() {
  digitalWrite(leftMotorForwardPin, HIGH);
  digitalWrite(leftMotorBackwardPin, LOW);
  digitalWrite(rightMotorForwardPin, HIGH);
  digitalWrite(rightMotorBackwardPin, LOW);
}
void goBackward() {
  digitalWrite(leftMotorForwardPin, LOW);
  digitalWrite(leftMotorBackwardPin, HIGH);
  digitalWrite(rightMotorForwardPin, LOW);
  digitalWrite(rightMotorBackwardPin, HIGH);
}
void stop() {
  digitalWrite(leftMotorForwardPin, LOW);
  digitalWrite(leftMotorBackwardPin, LOW);
  digitalWrite(rightMotorForwardPin, LOW);
  digitalWrite(rightMotorBackwardPin, LOW);
}
void goRight() {
  digitalWrite(leftMotorForwardPin, HIGH);
  digitalWrite(leftMotorBackwardPin, LOW);
  digitalWrite(rightMotorForwardPin, LOW);
```

```
  digitalWrite(rightMotorBackwardPin, LOW);
}
void goLeft() {
  digitalWrite(leftMotorForwardPin, LOW);
  digitalWrite(leftMotorBackwardPin, LOW);
  digitalWrite(rightMotorForwardPin, HIGH);
  digitalWrite(rightMotorBackwardPin, LOW);
}
```

6. Output / Observations

The **HC-SR04 ultrasonic sensor** continuously measures the distance between the robot and nearby obstacles. The Arduino UNO receives the distance information and compares it with a predefined safe distance. When no obstacle is detected, the robot moves forward. When an obstacle is detected within the specified distance, the Arduino sends control signals to the **L293D motor driver** to stop the motors, move backward, and then turn right to find another path.

1. The ultrasonic sensor continuously detects obstacles.
2. If no obstacle is detected, the robot moves forward.
3. If an obstacle is detected, the robot stops.
4. It then moves backward for a short time.
5. It then turns right to find another path.
6. The process continues automatically until the robot is turned off.

Observed Behavior:

Condition	Robot Behavior
No obstacle detected	Robot moves forward
Obstacle detected	Stops
Obstacle very close	Moves backward
After detecting obstacle	Turns right
No power	Motors stop

7. Result

The obstacle avoidance maze-solving robot was successfully designed and implemented using an **Arduino UNO, HC-SR04 ultrasonic sensor, L293D motor driver, and two DC motors**. The robot was able to detect obstacles in its path and change its direction automatically based on the detected distance.

The robot performed the required movements without manual control. **Therefore, the objective of the experiment was successfully achieved.**

8. Conclusion

In this experiment, I learned how to design and implement a robot using **Arduino UNO, HC-SR04 ultrasonic sensor, L293D motor driver, and DC motors**. The ultrasonic sensor was used to detect obstacles, while the Arduino processed the sensor data and controlled the motors through the L293D motor driver.

This experiment helped me understand the practical applications of **sensors, motor drivers, microcontrollers, and autonomous robot movement**. Overall, the experiment improved my understanding of basic robotic system design and Arduino programming.